

BCME Unit I

Complete Formula & Concept Sheet

Basic Civil & Mechanical Engineering — B.Tech 1st Year, 1st Semester

■ Unit: Basics of Civil Engineering

🔗 Exam-Optimised

📌 Flashcards + Tables + Quick Revision

1 Course Objectives

WHAT YOU WILL LEARN

- ▶ Role of Civil Engineering in modern society
- ▶ Importance of infrastructure development
- ▶ Civil engineering disciplines & their scope
- ▶ Properties & uses of construction materials
- ▶ Basics of building construction & planning

WHY IT MATTERS

- ▶ Foundation for all civil engineering courses
- ▶ Connects engineering to real-world infrastructure
- ▶ Helps understand material selection rationale
- ▶ Introduces sustainable & modern techniques
- ▶ Builds vocabulary for professional communication

2 Role of Civil Engineers in Society

INFRASTRUCTURE

- ▶ Roads, highways, bridges
- ▶ Railways & airports
- ▶ Dams & reservoirs
- ▶ Tunnels & underpasses

ENVIRONMENT

- ▶ Water supply & treatment
- ▶ Sewage & waste management
- ▶ Flood control systems
- ▶ Environmental impact assessment

HOUSING & ECONOMY

- ▶ Residential & commercial buildings
- ▶ Urban planning & development
- ▶ Employment generation
- ▶ GDP contribution via construction sector

Key Fact: Civil engineering is one of the oldest engineering disciplines. Civil engineers design, construct, supervise, operate, and maintain large construction projects.



STRUCTURAL ENGINEERING

- ▶ **Def:** Analysis & design of load-bearing structures
- ▶ **Scope:** Buildings, bridges, towers, dams
- ▶ **Key:** Load distribution, material strength, safety
- ▶ **App:** High-rise buildings, flyovers, stadiums



GEOTECHNICAL ENGINEERING

- ▶ **Def:** Study of soil & rock behaviour under structures
- ▶ **Scope:** Foundations, retaining walls, embankments
- ▶ **Key:** Soil testing, bearing capacity, settlement
- ▶ **App:** Foundation design, slope stability, tunnels



TRANSPORTATION ENGINEERING

- ▶ **Def:** Planning & design of transport systems
- ▶ **Scope:** Roads, railways, airports, harbours
- ▶ **Key:** Traffic flow, pavement design, safety
- ▶ **App:** National highways, metro rail, airports



HYDRAULICS & WATER RESOURCES

- ▶ **Def:** Management of water flow & resources
- ▶ **Scope:** Dams, canals, irrigation systems
- ▶ **Key:** Fluid mechanics, hydrology, water storage
- ▶ **App:** Hydropower, flood control, irrigation



ENVIRONMENTAL ENGINEERING

- ▶ **Def:** Protecting environment from human activity
- ▶ **Scope:** Water treatment, air quality, waste management
- ▶ **Key:** Pollution control, sanitation, sustainability
- ▶ **App:** Sewage plants, landfills, air pollution control

Discipline	Focus Area	Key Activities	Real-life Applications
Structural	Load-bearing structures	Analysis, design, safety assessment	Buildings, bridges, dams
Geotechnical	Soil & foundation	Soil testing, bearing capacity	Foundations, tunnels, embankments
Transportation	Movement of people & goods	Road design, traffic management	Highways, railways, airports
Hydraulics & WR	Water flow & management	Dam design, canal planning	Irrigation, hydropower, flood control
Environmental	Pollution & sustainability	Waste treatment, air quality	Sewage plants, landfills, EIA

STEPS IN BUILDING CONSTRUCTION

- ▶ 1. Site selection & survey
- ▶ 2. Soil investigation
- ▶ 3. Design & drawing preparation
- ▶ 4. Foundation construction
- ▶ 5. Superstructure (walls, columns, beams, slabs)
- ▶ 6. Roofing
- ▶ 7. Finishing (plastering, flooring, painting)
- ▶ 8. Services (plumbing, electrical)

PLANNING CONSIDERATIONS

- ▶ **Orientation:** Face building N-S for max sun & wind benefits
- ▶ **Ventilation:** Cross-ventilation; windows on opposite walls
- ▶ **Lighting:** Natural light via windows, skylights
- ▶ **Space Utilization:** Efficient room layout, no dead spaces
- ▶ **Safety:** Fire exits, structural stability
- ▶ **Accessibility:** Ramps, wide corridors

Types of Buildings:

Residential

Commercial

Industrial

Institutional

Mixed-use

6 Construction Materials

CEMENT **IMPORTANT**

- ▶ **Def:** Fine powder binding material; hardens with water
- ▶ **Properties:** High compressive strength, adhesive, durable
- ▶ **Uses:** Concrete, mortar, plastering
- ▶ **Adv:** Strong, versatile, readily available
- ▶ **Lim:** Poor tensile strength, cracks if not cured



AGGREGATES

- ▶ **Def:** Inert granular materials (sand, gravel, crushed stone)
- ▶ **Types:** Fine (sand <4.75mm), Coarse (>4.75mm)
- ▶ **Uses:** Filling in concrete & mortar
- ▶ **Adv:** Cheap, abundant, add bulk & strength
- ▶ **Lim:** Quality varies with source



BRICKS

- ▶ **Def:** Rectangular blocks of burnt clay
- ▶ **Properties:** Hard, durable, fire-resistant, porous
- ▶ **Uses:** Walls, partitions, pavements
- ▶ **Adv:** Economical, good insulation, locally available
- ▶ **Lim:** Heavy, absorbs water, poor tensile strength



CEMENT CONCRETE

- ▶ **Def:** Mixture of cement + water + fine & coarse aggregates
- ▶ **Properties:** High compressive strength, durable, rigid
- ▶ **Uses:** Foundations, slabs, beams, columns, roads
- ▶ **Adv:** Versatile, economical, fire resistant
- ▶ **Lim:** Low tensile strength (requires steel reinforcement)

⚙️ STEEL IMPORTANT

- **Def:** Alloy of iron + carbon; manufactured in mills
- **Properties:** High tensile & compressive strength, ductile, elastic
- **Uses:** RCC reinforcement, structural frames, roofing sheets
- **Adv:** Strong, recyclable | **Lim:** Costly, corrodes (needs protection)

7 Material Properties Comparison

Material	Compressive Strength	Tensile Strength	Durability	Cost	Primary Use
Cement	High (as concrete)	Low	High	Moderate	Binding agent
Aggregate	Moderate	Very Low	High	Low	Filler in concrete
Brick	Moderate	Very Low	Good	Low	Masonry walls
Concrete	Very High	Low	Very High	Moderate	Structural elements
Steel	Very High	Very High	High*	High	Reinforcement, frames

*Steel durability is reduced without corrosion protection (painting, galvanising, etc.)

8 Cement & Concrete — Key Details

TYPES OF CEMENT

- **OPC (Ordinary Portland Cement):** General purpose
- **PPC (Portland Pozzolana Cement):** Fly ash added; durable
- **Rapid Hardening Cement:** Sets quickly
- **Sulphate Resistant Cement:** Marine & soil exposure
- **White Cement:** Decorative purposes

CONCRETE INGREDIENTS

- **Cement:** Binding material
- **Water:** Hydration & workability
- **Fine Aggregate (Sand):** Fills voids
- **Coarse Aggregate (Gravel/Crushed Stone):** Bulk & strength
- **Admixtures (optional):** Improve specific properties

PROPERTIES OF GOOD CONCRETE

- **Workability** — easy to place & compact
- **Strength** — high compressive strength after curing
- **Durability** — resist weathering & chemical attack

CONCRETE MIX CONCEPT

- **Nominal Mix:** Fixed ratio (e.g. M20 = 1:1.5:3 — cement:sand:aggregate)
- **Design Mix:** Mix designed by testing for specific strength
- **W/C Ratio:** Lower = stronger (but less workable)

- Impermeability — low water permeability
- Economy — optimum use of materials

- **Curing:** Keeping moist for 28 days for full strength

Common Mix Grades: M10 (1:3:6) — Plain | M15 (1:2:4) — Mild | M20 (1:1.5:3) — Standard | M25 (1:1:2) — High strength

9 Prefabricated Construction

DEFINITION

Manufacturing of structural components (walls, floors, columns) at a factory off-site, then transporting and assembling them at the construction site.

✓ ADVANTAGES

- Faster construction
- Better quality control
- Less labour on site
- Less material wastage
- All-weather production

⚠ LIMITATIONS

- High initial cost
- Transport challenges
- Limited customization
- Skilled workers needed for assembly
- Joint failure risk

🏭 APPLICATIONS

- Low-cost housing
- Schools & hospitals
- Industrial buildings
- Modular offices
- Bridge decks

10 Traditional vs Prefabricated Construction

Parameter	Traditional Construction	Prefabricated Construction
Speed	Slow — sequential processes	Fast — parallel production
Cost	Lower initial cost	Higher initial; lower long-term
Quality	Site-dependent; variable	Controlled factory conditions
Flexibility	High — custom design possible	Limited — standardized units
Labour	High site labour	Less site labour; factory workers
Weather Impact	Highly affected	Factory unaffected; site assembly affected
Wastage	High material waste	Minimal — precise cutting

★ Cement vs Concrete vs Steel

Feature	Cement	Concrete	Steel
Type	Binding powder	Composite material	Metal alloy

Feature	Cement	Concrete	Steel
Comp. Strength	High (in paste)	Very High	Very High
Tensile Strength	Very Low	Low	Very High
Weight	Light (powder)	Heavy (~2400 kg/m³)	Very Heavy (~7850 kg/m³)
Cost	Moderate	Moderate	High
Corrosion	No corrosion	No (protects steel inside)	Corrodes without protection
Use	Binder in concrete/mortar	Structural elements	Reinforcement, frames

Q: What is Civil Engineering?

A: Branch of engineering dealing with design, construction & maintenance of infrastructure

Q: What is Structural Engineering?

A: Analysis & design of load-bearing structures

Q: What is Geotechnical Engineering?

A: Study of soil & rock behaviour for foundations

Q: What is Transportation Engineering?

A: Planning & design of roads, railways & airports

Q: What is Environmental Engineering?

A: Protecting environment from pollution & human activity

Q: What is Hydraulics Engineering?

A: Management of water flow, dams & irrigation systems

Q: What is cement?

A: Fine powder binding material that hardens with water

Q: Full form of OPC?

A: Ordinary Portland Cement

Q: Full form of PPC?

A: Portland Pozzolana Cement

Q: What are aggregates?

A: Inert granular materials (sand, gravel, stone) used as filler in concrete

Q: Fine aggregate size?

A: Less than 4.75 mm (sand)

Q: Coarse aggregate size?

A: Greater than 4.75 mm (gravel/crushed stone)

Q: What is concrete?

A: Mixture of cement + water + fine & coarse aggregates

Q: M20 concrete mix ratio?

A: 1:1.5:3 (cement:sand:aggregate)

Q: What is W/C ratio?

A: Water-Cement ratio; lower = stronger concrete

Q: Curing period of concrete?

A: 28 days for full strength

Q: What is steel?

A: Alloy of iron + carbon with high tensile strength

Q: Density of concrete?

A: ~2400 kg/m³

Q: Density of steel?

A: ~7850 kg/m³

Q: What is a brick?

A: Rectangular block of burnt clay used for masonry

Q: What is prefabrication?

A: Manufacturing components off-site in factory & assembling on-site

Q: Main advantage of prefabrication?

A: Speed & quality control

Q: Main limitation of prefabrication?

A: High initial cost & limited customization

Q: Best building orientation?

A: North-South for maximum ventilation & natural light

Q: What is workability of concrete?

A: Ease with which concrete can be placed & compacted

Q: What is RCC?

A: Reinforced Cement Concrete — concrete with embedded steel

Q: Weakness of concrete?

A: Low tensile strength

Q: Weakness of steel?

A: Corrosion (rusting) without protection

Q: What is bearing capacity?

A: Maximum load soil can support without failure

Q: What is a nominal mix?

A: Fixed ratio concrete mix (e.g. M20 = 1:1.5:3)

! Exam Tips & Common Mistakes

⚠ MISTAKE 1

Confusing civil engineering disciplines

Remember: Structural = structures | Geotechnical = soil | Transportation = movement | Hydraulics = water | Environmental = pollution

⚠ MISTAKE 2

Mixing up cement and concrete

Cement is a component; Concrete is the final product. Don't use the words interchangeably in answers.

⚠ MISTAKE 3

Forgetting tensile vs compressive strength

Concrete is strong in compression but weak in tension. Steel is strong in BOTH. RCC uses steel to compensate for concrete's weakness.

⚠ MISTAKE 4

Writing long explanations

Examiners prefer keywords + bullet points + tables. A well-drawn table can earn full marks faster than 5 paragraphs.

⚠ MISTAKE 5

Ignoring mix ratios & grades

Remember M10, M15, M20, M25 grades and their nominal mix ratios — commonly asked in 2-mark questions.

⚠ MISTAKE 6

Missing advantages & limitations

For any material question — always include both advantages AND limitations. Incomplete answers lose marks.

Pro Tip: Use comparison tables in answers whenever asked to compare two items (e.g., traditional vs prefab, cement vs steel). Examiners love structured answers.

⚡ Quick Revision Sheet — Last 15 Minutes

5 Disciplines

Structural

Geotechnical

Transportation

5 Materials

Cement

Aggregates

Bricks

Concrete

Hydraulics

Environmental

Steel

Key Definitions

OPC

PPC

RCC

W/C Ratio

Curing = 28 days

Prefabrication

Bearing Capacity

Nominal Mix

Mix Grades to Remember

M10 → 1:3:6

M15 → 1:2:4

M20 → 1:1.5:3

M25 → 1:1:2

🔑 ONE-LINERS TO MEMORISE

- ▶ Concrete = Cement + Water + Fine agg. + Coarse agg.
- ▶ Cement: High comp. strength | Low tensile strength
- ▶ Steel: High tensile + compressive strength | Corrodes
- ▶ Brick: Burnt clay | Good insulation | Low tensile strength
- ▶ Prefab: Factory-made → assembled on site
- ▶ Curing: Keeps moist for 28 days for full strength
- ▶ W/C ratio: Lower → Stronger (but less workable)
- ▶ Building orientation: North-South = Best